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Short communication

Chemically recyclable rosin-based polymers

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ABSTRACT

The pursuit of next-generation sustainable polymers with chemical recyclability and resource renewability has become a priority in materials science. Rosin, a widely available renewable natural biomass, holds promise in this regard. In this study, dehydroabietic acid was incorporated into a cyclooctene monomer structure as pendant groups. This tailored monomer with low ring strain possesses the ability to efficiently undergo ring-opening metathesis polymerization (ROMP), resulting in the synthesis of rosin-based unsaturated polymers with a fully hydrocarbon backbone. The molecular weight of these polymers, ranging from 50 kDa to 160 kDa, can be easily modulated by varying the monomer-to-catalyst feeding ratio. Remarkably, these amorphous polymers exhibit exceptional thermostability, with a decomposition temperature exceeding 400 °C. This type of polymer exhibits excellent acid and alkali resistance within a pH range of 3 to 11. Upon exposure to Grubbs catalyst under mild conditions, these polymers undergo depolymerization back to the native monomer in a highly efficient manner, achieving a remarkable recovery ratio of approximately 95 %. It is evident that this innovative approach is not only relevant for the development of rosin-based polymers but also holds promise for informing the design of other renewable polymers with superior thermostability and enhanced chemical recyclability.

1. Introduction

Given the increasing concerns about the health and environmental effects of plastic waste, creating a circular plastics economy appears to be the most effective solution.[1–4] In this model, sustainability considerations would dictate the full life cycle of plastics use, including replacing petrochemical plastics with biorenewable plastics.[5,6].

Tall oil, sourced abundantly from forestry products, offers cost-effective production potential for future endeavors in sustainability development.[7] Annually, between 1.6 and 2 million tonnes of crude tall oil are produced, with rosin acid comprising 30 % to 60 % of the main compound groups.[8] Rosin acid, notably abietic acid, is characterized by hydrophenanthrene rings.[9] Given its renewability and stable rigid structure, rosin acid has the potential to supplant petrochemical aromatic ring raw materials in the modification and preparation of fine chemicals and polymer materials.[8] There has been a focus on utilizing rosin derivatives as renewable monomers in various polymerization processes, including free radical polymerization,[10] step-growth polymerization,[11–17] controlled radical polymerization,[18–24] ring-opening polymerization,[18,25] or post-polymerization modification[26] to develop rosin-containing thermoplastic polymers.

Although rosin-based polymers have significantly contributed to the sustainable polymer family, not all of them are degradable.[26] For instance, rosin-based polymers prepared through radical polymerization have chemical inert carbon–carbon bonds in their backbone, rendering them challenging to degrade.[27,28] The incorporation of degradable functionalities into these polymer backbones during step-growth polymerization or post-polymerization modification is crucial for reducing environmental persistence.[6,26,29–31] However, this modification often leads to a compromise in the stability and properties of rosin-derivative polymers during storage or service.[32,33] As a result, there is a substantial challenge in balancing the trade-off between degradability and the desired material properties of rosin-derivative polymers.[34,35] In the transition to a circular bio-plastics economy, future generations of plastics should focus on bio-advantaged designs that prioritize high efficiency in chemical recycling for monomer recovery at the end of life. To the best of our knowledge, chemically recyclable rosin-based polymer has not been reported yet.

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By selecting non-hydrolytic bonds as the polymer backbone that are inherently stable but reactive to specific chemical stimuli for recycling, the dilemma mentioned can be effectively addressed.[36] For instance, carbon–carbon double bonds possess high bond dissociation energy, rendering them resistant to acid and alkali environments; however, they are sensitive to ruthenium carbene catalysts.[37] The Wang Group has recently introduced a range of cyclooctene-based polymer systems designed for chemical recycling by reducing the ring strain energy of monomers.[38,39] Demonstrating the robustness of this approach, our recent research revealed the successful installation of a *trans*-dioxolane ketal at the 5,6-positions of cyclooctene, incorporating lignin derivatives as pendants.[40] This strategy could potentially be extended to incorporate rosin derivatives, allowing for further customization of the properties of the chemically recyclable parent polymer template. Additionally, the integration of molecular biomass into the chemically recyclable polymer skeleton enhances the renewability of these polymers, highlighting their environmentally sustainable characteristics.

In this study, we detailed the synthesis of a low ring strain cyclooctene monomer derived from dehydroabietic acid for producing polyolefins with rosin side chains via ring-opening metathesis polymerization (ROMP). The thermal characteristics, acid/base stability, and chemical recyclability of the resulting materials were examined.

2. Results and discussion

2.1. Monomer synthesis and characterization

After reduction into **1** using lithium aluminium hydride, the commercial available dehydroabietic acid can be purified, removing all impurities through column chromatography with moderate yield. The reduced product, as shown in Scheme 1, undergoes a reaction with 4-oxocyclohexane-1-carboxylic acid to yield the precursor ester **2**. Subsequently, the target monomer **3** is prepared through acetalization with *cis*-cyclooctene-*trans*-5,6-diol. This reaction route boasts a total yield exceeding 45 %. The structures of all monomers were meticulously verified using ^{1}H NMR, ^{13}C NMR, and high-resolution mass spectrometer (HRMS, Supporting Information). Characteristic signals of dehydroabietic acid-derived precursor and monomer structure are illustrated in Fig. 1, with confirmation based on the chemical shifts of methylene, aromatic, and vinyl protons. Notably, all integrations of proton NMR spectra correlate closely with the theoretical structures, underscoring their exceptional purity level (Figure S4, S6 and S8).

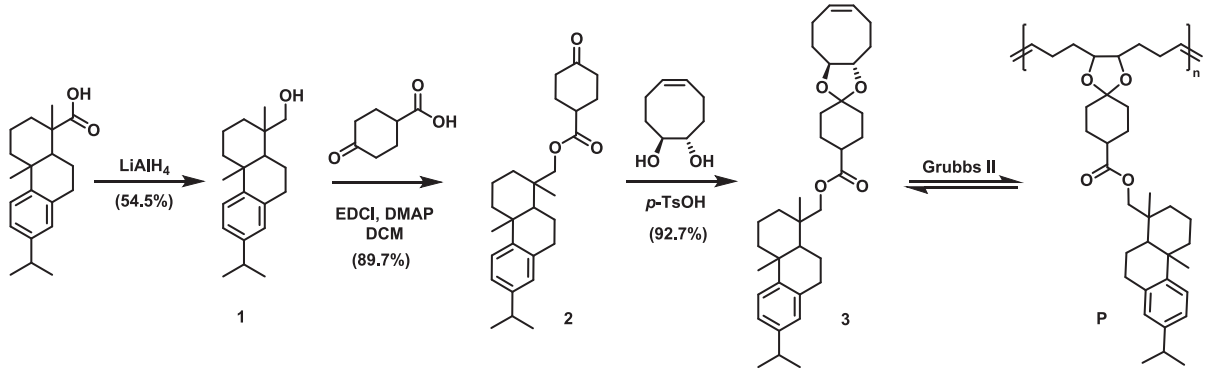
2.2. Synthesis and structural characterization of polymers

The Grubbs II catalyst allows for the room temperature ROMP of the cyclooctene-based monomer in dichloromethane. After polymerization, the residual catalyst in the polymer can be completely removed by adsorption using Quadrapure TU macroporous particles to ensure the purity of the polymer. A high monomer concentration of 2.0 M is preferred for promoting the polymerization process.[38,39] Initially, a monomer-to-initiator feeding ratio of 500:1 was used in the first experiment, resulting in the formation of **P1** with a molecular weight exceeding 50000 Da. Fig. 2 illustrates the gel permeation chromatography (GPC) traces of **P1**, showing that as the monomer-to-initiator feeding ratio was increased, the elution peak shifted towards higher molecular weights. Table 1 presents a summary of the detailed results. The ^{1}H NMR spectra displayed in Fig. 1 revealed the disappearance of characteristic signals of vinyl protons at 5.6 ppm in **P1**, with new peaks appearing at 5.3–5.5 ppm, indicating the presence of ring-opened units in the unsaturated polymer backbone. Notably, the signals corresponding to the rosin-derived pendant groups remained largely unchanged after polymerization when compared to those of the monomer.

2.3. Thermal properties

The thermal properties of **P1**-**P4** were characterized using differential scanning calorimetry (DSC). All four polymers were identified as non-crystalline plastics with glass transition temperatures (T_g) falling within the range of 70–90 °C, as illustrated in Fig. 3. The observed T_g values were found to be influenced by the molecular weight of the polymers, attributable to the chain end effects. Unlike the lignin-based mimics reported earlier, rosin-based polymers showed higher T_g values at comparable molecular weights.[40] The high T_g of rosin-containing polymers is mainly due to the bulky side groups rather than its polarity. The polymer derived from dehydroabietic acid is expected to have weaker intermolecular interactions compared to a lignin-based polymer, which contains more oxygen-containing groups and therefore exhibits greater polarity. The pendant dehydroabietic acid in the polymer contributes to its large size, thus elevating the energy needed for internal bond rotations. As a result, the polymer chain becomes rigid and restricts chain segmental mobility, ultimately leading to an increase in T_g.

The thermal stability of **P1**-**P4** was analyzed and compared using thermogravimetric analysis (TGA) under nitrogen (Fig. 4). TGA curves for these rosin-based materials exhibited similar single-stage weight loss behavior, with $T_{d,5\%}$ up to 400 °C and $T_{d,max}$ up to 442 °C. **P1**, having a low molecular weight and the highest concentration of chain end groups, shows lower thermal stability compared to the other samples (Table 1). The concentration of end groups impacts the decomposition temperature of the polymer, particularly influencing the $T_{d,5\%}$ value in the early stages of decomposition.[19,41,42] The disparity in $T_{d,5\%}$ values among **P1**-**P4** with varying molecular weights can reach as much as 30 °C. Conversely, the $T_{d,max}$ is primarily determined by the inherent bond dissociation energy of the polymer chain.[43] As a result, the $T_{d,max}$ values for **P1**-**P4** with different molecular weights exhibit minimal variation, typically within 5 °C (Table 1). The high thermal decomposition temperatures of **P1**-**P4** were found to be comparable to



Scheme 1. Synthesis of rosin-based monomer and polymer.

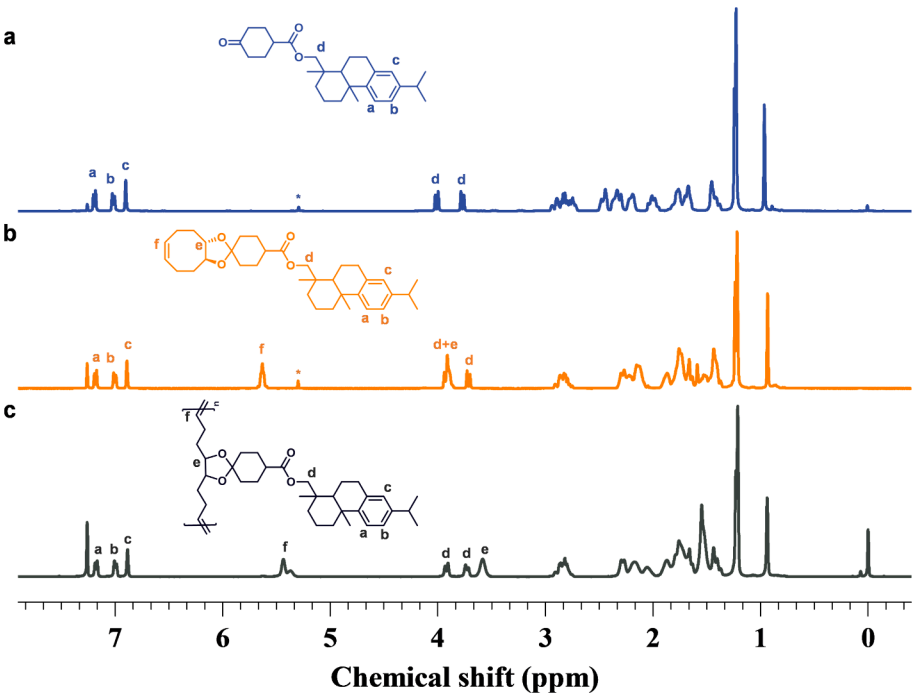


Fig. 1. ^{1}H NMR (400 M, 25 °C, $CDCl_3$) spectra of (a) **2**, (b) **3**, and (c) **P1**.

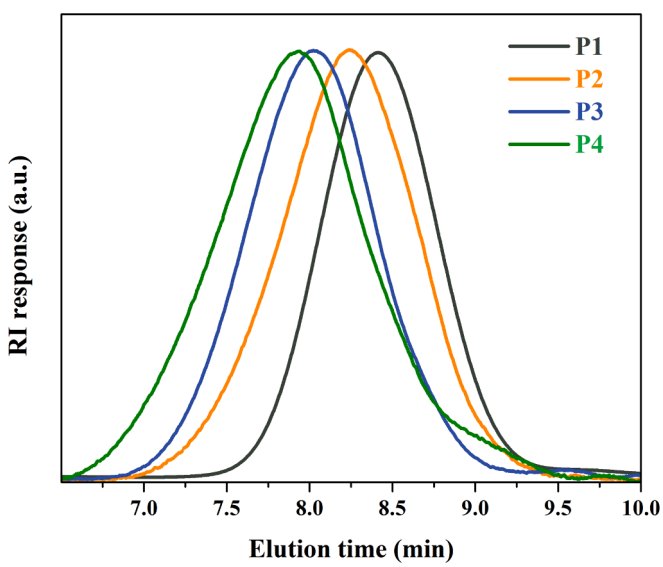


Fig. 2. GPC traces of **P1**-**P4**.

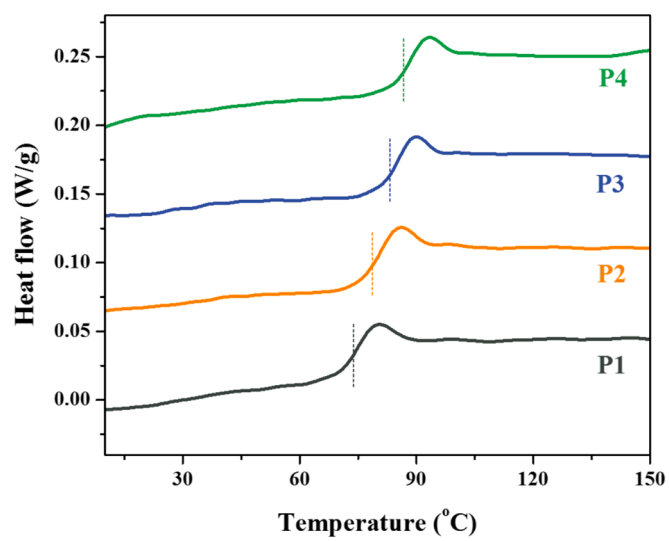


Fig. 3. The second heating scan of DSC curves for **P1**-**P4**.

Table 1
Results of ROMPs of monomer **3** catalyzed by Grubbs II.

Entry	[M]: Cat.	M_n [a]	$Đ$ [a]	*trans:cis* [b]	T_g [c] (°C)	$T_{d,5\%}$ [d] (°C)	$T_{d,max}$ [e] (°C)	Depolymerization ratio [f]
P1	500: 1	52,500	1.80	70:30	73.5	368.5	437.6	93 %
P2	600: 1	73,400	2.05	77:23	78.5	379.8	438.1	97 %
P3	1500: 1	135,700	1.96	74:26	83.5	399.6	442.4	96 %
P4	2000: 1	160,600	2.23	70:30	86.8	397.5	442.7	95 %

[a] Determined by GPC in THF relative to narrow-distributed polystyrene standards.
[b] Determined by the integration of *trans/cis* olefinic peaks of polymers in ^{1}H NMR spectra.
[c] Determined by DSC during the second heating scan via a STARe method (10 K/min).
[d] Defined as the temperature at which 5% weight loss occurred under N_2 atmosphere.
[e] Defined as the temperature at which fast weight loss occurred under N_2 atmosphere.
[f] Determined by the integration of olefinic peaks from polymer and monomer in ^{1}H NMR spectra.

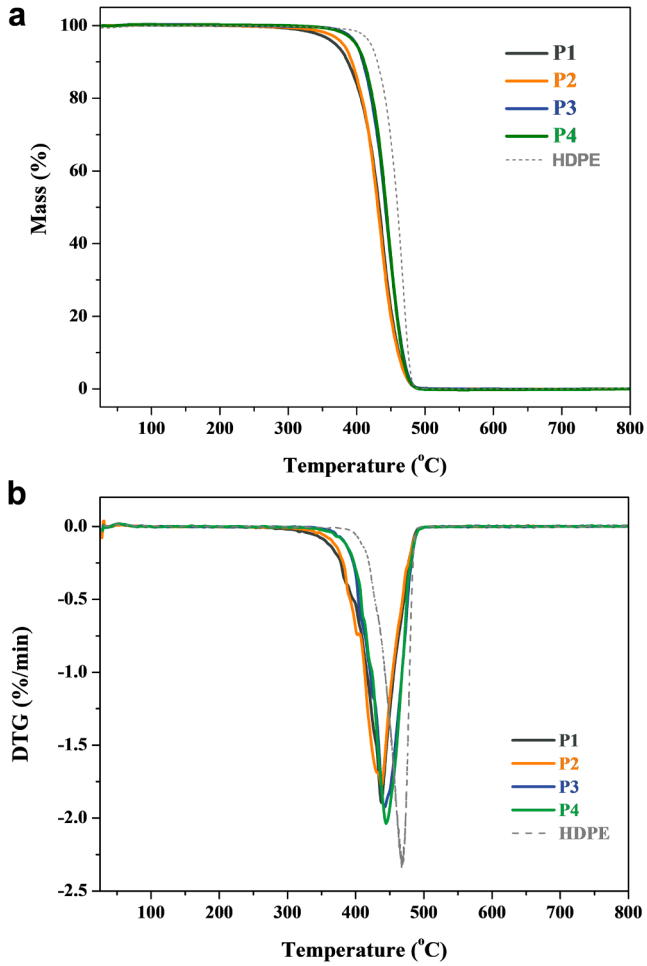


Fig. 4. Thermal gravimetric analysis (TGA) thermograms for HDPE and **P1**-**P4**.

commercial HDPE, suggesting a noteworthy thermostability of these polymers. This exceptional thermostability is likely attributed to the all-hydrocarbon backbone nature of **P1**-**P4**, which is consistent with previous findings on rosin-based polymers synthesized with similar backbones through radical polymerization or ROMP.[18,25,44] On the other hand, rosin-based polymers derived from step-growth polymerization, featuring labile chemical bonds, tend to exhibit lower thermostability compared to those with an all-hydrocarbon backbone structure.[11–13,15,17,45].

2.4. Mechanical properties

It is essential to highlight the significance of mechanical properties as crucial parameters for polymer materials. **P1**, which has the lowest molecular weight (M_n = 52.5 kDa), is a fine white powder that cannot be molded (Figure S12). Despite increasing the molecular weight of **P4** to 160 kDa, the specimens remain too brittle for effective demolding towards mechanical testing (Figure S12). This brittleness issue is commonly encountered in rosin-based polymers due to their exceptionally high chain entanglement molecular weight (M_e). Tang's research revealed a M_e value of up to 86 kDa for a rosin-based polynorbornene.[18] Given that the side-chain group in our study is even larger than the rosin-based polynorbornene, we anticipate that the M_e of our recyclable rosin-based polymer will exceed 86 kDa. Consequently, rosin-based plastics (**P1**-**P4**) with molecular weights lower than or comparable to M_e exhibit inadequate chain entanglements and long-range intermolecular interactions, leading to brittle behaviors. Further enhancing the molecular weight proves challenging at present.

2.5. Acid/base stability

To assess the hydrolytic stability of the ronsin-based polymer, we conducted experiments under various conditions known to affect ester and ketal bonds differently. Ester bonds tend to be unstable in alkaline solutions,[46] while ketal bonds are susceptible to degradation in acidic environments.[47] In the experiment using **P4** as the subject, small pieces of **P4** were immersed in aqueous solutions with varying pH levels. The test conditions included mild acidic solution (pH = 5.5), strong acidic solution (pH = 3), mild alkaline solution (pH = 8.5), and strong alkaline solution (pH = 11). The potential degradation pathway is outlined in Scheme 2, emphasizing labile bonds in red. If small molecules dissociate from the polymer backbone, the solution will become cloudy. The results showed that all solutions remained clear after seven days (refer to Figure S13), suggesting the absence of significant hydrolysis of either the ketal or ester bonds. Furthermore, the small polymeric pieces were observed to remain intact after the 7-day period, indicating the structural integrity of the polymer backbone.

The ^{1}H NMR spectra (Figure S14) of the polymer sample exhibit no difference after treatment with acid/base solutions, and protons signals indicating the formation of ketones or alcohols were not detected. Additionally, further GPC elution traces (Figure S15) revealed no signs of degradation. Therefore, it can be concluded that both the backbone and linker bonds of the polymer are robust under acid and alkali environments. The cyclic ketal structure of **P4** provides excellent acid resistance due to the torsional strain from the substituted ketal groups.[47].

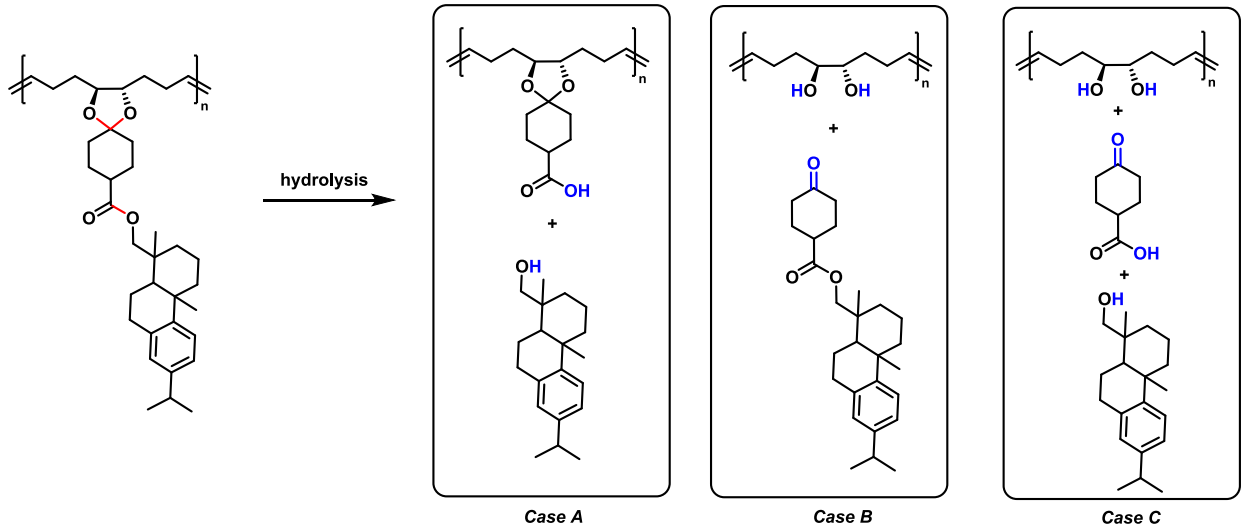
2.6. Depolymerization studies

The low ring strain energy of the monomer, theoretically calculated as 5.1 kcal/mol by determining the enthalpy change ΔH through the homodesmotic reaction shown in Fig. 5,[39,40] contributes to the formation of a polymer with a relatively low ceilingg temperature (T_c). This characteristic of the monomer results in a polymerization process that favours depolymerization at elevated reaction temperatures and low concentrations.[36,48].

The depolymerization of **P1** was conducted at a concentration of 20 mM with Grubbs II (1 % mol) as the catalyst under gentle heating (50 °C) for 2 h. In contrast, the polymerization was carried out at a concentration of 2 M at 25 °C. Fig. 6a displays the GPC elution curves, which reveal that the parent polymer peak was completely eliminated after depolymerization. Notably, a new dominant peak emerged in the depolymerization system, showcasing a molecular weight that closely resembled that of the monomer. The comparison of elution time with the pristine monomer confirmed that this dominant peak in the depolymerization system corresponds to the recovered monomers. Moreover, a shoulder peak in the depolymerization system, with molecular weights consistent with a dimer, implies an equilibrium between cycles of different ring-sizes after depolymerization.[49–51] Upon work-up of the collected depolymerized solid mixtures, a neat monomer with a yield of 93 % was recovered. Proton NMR spectra confirmed that the quality of the recycled monomer matched that of the pristine monomer (Fig. 6b). Similarly, other polymers (**P2**-**P4**) of varying molecular weights also underwent efficient ring-closing depolymerization, resulting in the recovery of monomers with a high recovery ratio of approximately 95 %. This observation suggests that the recovery ratio is independent of the molecular weight.

3. Conclusions

In summary, a low-strained rosin-based cyclooctene monomer was developed for ROMP. The resulting polymer, featuring rosin as pendants, demonstrates exceptional acid/base stability (pH = 3–11) and thermal stability, as evidenced by decomposition temperatures ($T_{d,5\%}$) reaching up to 400 °C. Depolymerization yields were as high as 95 %,



Scheme 2. Possible degradation pathway of rosin-based polymers.

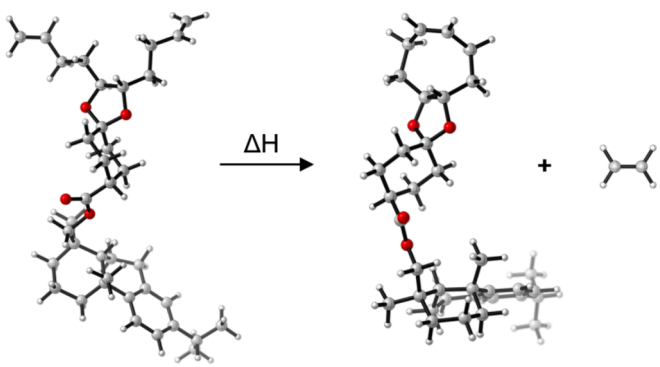


Fig. 5. Geometrical structure of monomer before and after ring-opening.

with nearly complete recovery of the monomer achievable under relatively mild conditions. Looking ahead, the incorporation of degradable motifs into the polymer backbone is crucial for enabling chemical recyclability. However, it is important to note that while this modification enhances recyclability, it may also compromise the stability of materials during storage due to unintentional chain rupture. This novel platform, characterized by an all-hydrocarbon backbone, enables precise spatiotemporal control over degradability but exhibits selectivity towards the Grubbs catalyst in specific conditions. Consequently, this approach offers a potential solution to the challenges posed by the trade-offs between stability and depolymerizability. We anticipate that additional molecular biomass with novel functionalities can be integrated into this robust platform in the near future.

CRediT authorship contribution statement

Ye Sha: Writing – review & editing, Conceptualization. **Xiaofan Chen:** Methodology, Investigation, Data curation. **Wei Sun:** Methodology, Investigation. **Yuan Hu:** Validation, Conceptualization. **Chengcheng Wang:** Data curation. **Enhua Xu:** Software. **Zhenyang Luo:** Supervision. **Puyou Jia:** Writing – review & editing, Supervision.

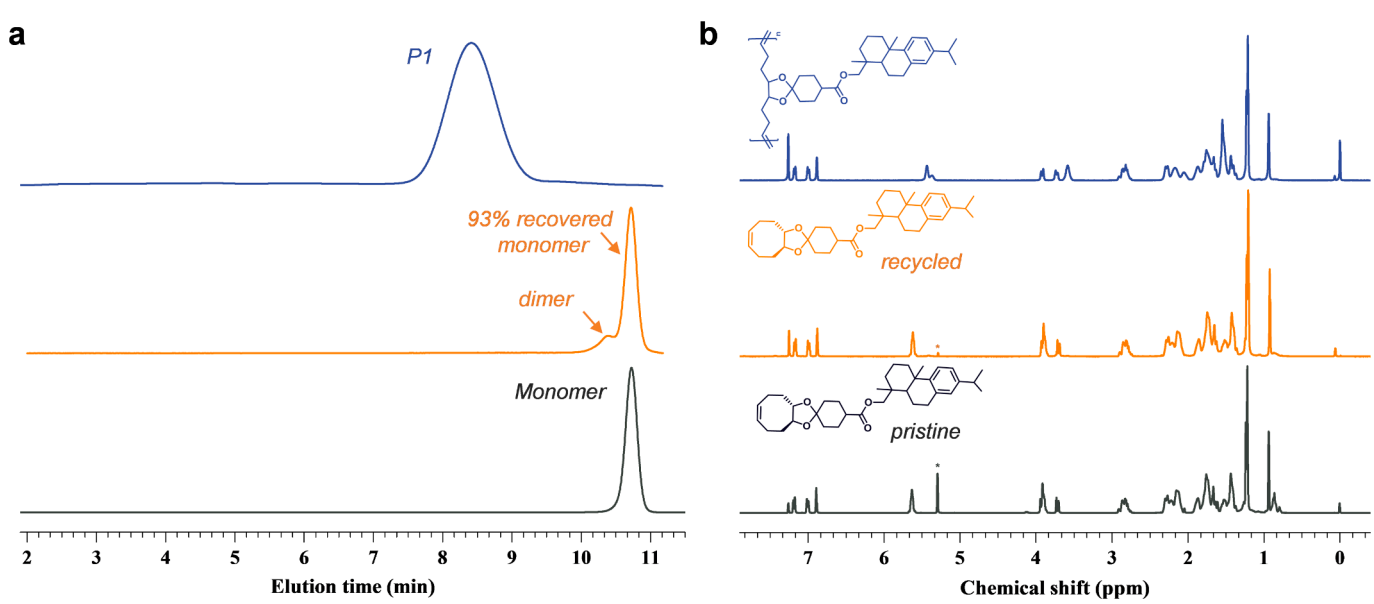


Fig. 6. (a) GPC elution straces of **P1**, depolymerized system and starting monomer. (b) ^{1}H NMR (400 M, 25 °C, $CDCl_3$) spectra of **P1**, recovered monomer and starting monomer.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113141.

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